

Selecting the Right Wireless Module



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There has been a significantly increased interest in the wireless and radio frequency (RF) modules in research as well as industry, feel industry experts. Alok Jain, co-founder, InkOcean, in, says, "It is primarily being driven by the hottest technology buzz of today—the Internet of Things (IoT). There are claims of many trillion dollars of economic output in the next few years in the realm of IoT devices."

One of the most important elements that help make devices IoT-enabled is their being wirelessly connected to other devices or the device hub. Jain adds, "Over the past couple of years, many low-power wireless protocols have been jostling for this space. Noteworthy technologies among these are Wi-Fi, Bluetooth Low Energy (BLE), ZigBee and RF."

Multiple wireless technologies such as Wi-Fi, ZigBee, near field communication (NFC), RF, BLE and IPv6 over low-power wireless personal area networks (6LoWPAN), to name a few, are integrated in modules today. Praveen Ganapathy, director applications-processors, sales and applications, Texas Instruments (TI) India, informs, "At a macro-level, considering all these technologies, today's modules have low power, high bandwidth, high data throughput, high-level security and certifications. There is a trend towards integra-

tion of microcontrollers in modules as well."

He adds, "Traditionally, RF devices were separate components that needed separate processors or controllers, but integration of low-power microcontrollers with advanced sleep/power save modes enable wearable devices."

Modules today also have the capability to handle RF in different bands. Ganapathy says, "Nowadays, there is higher transmission in 2.4GHz range, resulting in network congestion. There is a trend towards using sub 1GHz and 5GHz bands to achieve a higher range."

IoT-driven modules

IoT is driving the usage of wireless modules in a big way. Higher throughput and better performance are not the major factors looked at when buying a module, it is the high level of integration in these, informs Dhiraj Sogani, general manager and senior vice president, System BU, Redpine Signals Inc. Integration of multiple wireless standards is very critical for the IoT market. He says, "It is difficult to find a single wireless technology that will meet everyone's requirements. Each has its own niche and application."

Today's RF modules for wireless applications include options for dual-band and quad-band and support key air interfaces including 2G, 3G, 4G (LTE), ZigBee, Wi-Fi and WiMax, which are favourable for IoT applications, informs Abhishek Rao, technical marketing manager, element14. He says, "The latest addition features an ultra-small core, which enables a much smaller footprint than the previous generation, delivering significant size reduction and greater platform flexibility across applications and end products." He adds, "By incorporating the latest modules, IoT applications will have ultra-long ranges with minimal power requirement, boosting the overall system performance."

IoT is a horizontal market, and fields like industrial, medical, automotive, wearable or healthcare are some of the verticals where wireless modules are used. Sogani explains,

Some latest modules for your next IoT application



Atmel's WILC3000/WINC1500
series of Wi-Fi solutions



Kneron's
SmartWiFi



Microchip's RN2483
433MHz/868MHz module



RedBearLabs' BLE Mini -
Bluetooth 4.0 Low Energy
module



Redpine Signals' RS9113
WiSeConnect module



Texas Instruments' SimpleLink
Wi-Fi CC3200 LaunchPad



Texas Instruments' WL18x7M00
dual-band combo 2x MIMO Wi-Fi modules